

Forecasting the S&P 500 Index Using Recurrent Neural Networks

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1. Introduction

The stock market plays a pivotal role in the global economy, and the S&P 500 Index is widely regarded as a key benchmark for U.S. economic performance. As a composite of 500 leading publicly traded companies, its daily movements capture investor interest, economic outlook, and market trends. Given its significance, accurately forecasting the S&P 500's behavior holds substantial value for traders, financial analysts, economists, and policymakers.

This project, conducted by Team Neural Nexus for the ADTA 5560 course, explores the potential of Recurrent Neural Networks to model and forecast the daily closing prices of the S&P 500 Index. Leveraging over 40 years of historical financial data sourced from Yahoo Finance, the team aims to test how well RNNs capture time-based patterns in stock data. Through this work, the team seeks to uncover patterns that could enhance market predictability and support informed decision-making in the financial sector. In addition to assessing the predictive power of RNNs, the project investigates the impact of varying look back windows. The insights from this project may offer practical applications in trading strategies, portfolio management, and economic forecasting.

2. Literature Review

Forecasting stock markets is a difficult process because prices follow complicated, constantly changing patterns. Recent advances in RNNs and other deep learning models such as, Long Short-Term Memory (LSTM) models have shown promise at handling these patterns. Fischer and Krauss (2018) found that LSTMs beat older methods like Random Forests and Support Vector Machines when predicting movements in the S&P 500. Their key advantage is remembering trends over long periods, which makes RNNs a natural choice for our project of modeling daily closing prices overtime.

Krauss, Do, and Huck (2017) compared deep neural networks, gradient-boosted trees, and random forests, showing that when deep models are properly set up and fed the right inputs like opening price, trading volume, and daily highs and lows they often make more accurate predictions. Zhang and Zheng (2019) focused specifically on LSTMs for short-term forecasting and showed they can pick up useful patterns from recent data. Their work suggests that using a “look-back” window of recent days can help fine-tune how far ahead the model should predict, guiding our choices on window size for both one-day and week-ahead forecasts.

More recent reviews, like Shah, Vaidya, and Shah (2022), highlight how mixing different methods such as combining ARIMA with LSTM or adding convolutional layers before a bidirectional LSTM can make predictions more stable when markets swing wildly. Although our current focus is on standalone RNN models, these hybrid ideas point toward future improvements. They also stress the importance of clear error measures for example, RMSE and MAPE, which helps us set up a solid framework for testing and comparing our models' performance.

3. Research Questions

Our Research aims to use RNN's to answer the following questions:

1. Can Recurrent Neural Networks (RNNs) effectively model and forecast the S&P 500's daily closing prices?
2. How do different look-back windows (e.g., 5, 20, or 60 trading days) influence model performance and accuracy?
3. How accurately can the model forecast short-term versus medium-term price movements?
4. Can historical stock market data be used to predict long-term trends in the S&P 500, such as 1-year, 5-year, or 10-year closing price forecasts?

4. EDA/ Data Preparation

4.1.1 Importing Libraries

The analysis begins by importing standard Python libraries:

- **Pandas** and **NumPy** for data handling
- **Matplotlib** and **Seaborn** for visualization
- **SciPy.stats** for statistical functions
- **scikit-learn** tools (a RandomForestRegressor model and LabelEncoder) for the feature-importance analysis.

These libraries provide data structures (e.g. DataFrames), plotting utilities, and machine learning algorithms needed for the EDA.

4.1.2 Loading the Dataset

The dataset (a CSV file named Yahoo dataset.csv, containing daily stock data) was loaded into a Pandas DataFrame using `pd.read_csv`. The first five rows were printed, revealing columns **Date**, **Close**, **High**, **Low**, **Open**, and **Volume**. This confirms that the data was successfully read and contains the expected fields.

4.1.3 Initial Data Inspection

After loading, the data was inspected for structure and cleanliness:

- **First Few Records (head):** The `head()` output shows initial entries such as `Date=4/20/1982, Close=115.44, High=117.14, Low=114.83, Open=115.80, Volume=54610000`. This confirms the format of each record (daily OHLC prices plus volume).
- **Data Info:** Using `data.info()`, the DataFrame has **10,765 rows** and 6 columns. All columns have 10,765 non-null entries, meaning **no missing values**. The Date column is of type *object* (string) and the others are numeric (float or int).
- **Missing Values:** A check of `data.isnull().sum()` shows **0 missing values** in every column, indicating the dataset is complete with no NaNs to handle.

Together, these steps confirm that the dataset is fully loaded, with consistent schema and no missing entries.

4.1.4 Descriptive Statistics

Descriptive statistics were computed via `data.describe()`, summarizing each numeric feature (Close, High, Low, Open, Volume).

Statistic	Close (float64)	High (float64)	Low (float64)	Open (float64)	Volume (int64)
Count	10,765	10,765	10,765	10,765	10,765
Mean	1427.33	1435.27	1418.26	1427.04	2,071,578,000
Std. Dev	1267.22	1273.52	1260.17	1267.08	1,901,987,000
Min	102.42	103.01	102.20	102.42	14,990,000
25% (Q1)	431.04	431.93	429.20	430.95	234,700,000
Median (Q2)	1143.22	1150.30	1135.45	1143.22	1,430,900,000
75% (Q3)	1871.89	1878.28	1859.79	1872.28	3,659,820,000
Max	6090.27	6099.97	6079.98	6089.03	11,456,230,000

Table 4.1.4 describes the central tendency, dispersion, and shape of the distribution.

Overall, the descriptive stats indicate that the price variables are broadly distributed with significant spread, and the volume data is also highly variable. The right-skew in each distribution (mean > median) is evident from the percentiles.

4.1.5 Outlier Detection

Outliers were identified using the Z-score method on all numeric features. Any value with $|z\text{-score}| > 3$ was flagged as an outlier. This revealed the following counts of outliers in each column:

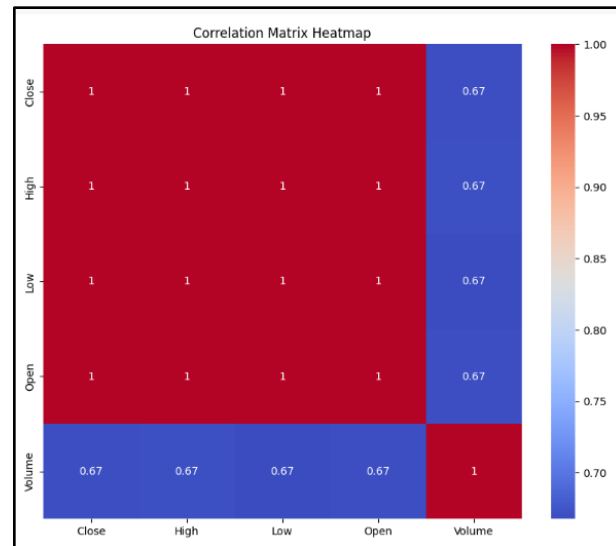
- **Close:** 163 outliers
- **High:** 162 outliers
- **Low:** 165 outliers
- **Open:** 164 outliers
- **Volume:** 56 outliers

These numbers correspond to roughly 1–1.5% of the price observations and about 0.5% of volume, indicating that a small but notable fraction of days have extreme values. These outliers in the price columns are consistent with the very wide ranges and heavy tails seen in the descriptive statistics. In practice, these outliers likely reflect unusually high or low market swings on certain days.

4.1.6 Correlation Analysis

A correlation matrix was computed for the numeric features. The result (below) shows **very high correlation** (~ 0.999) among the price-related columns:

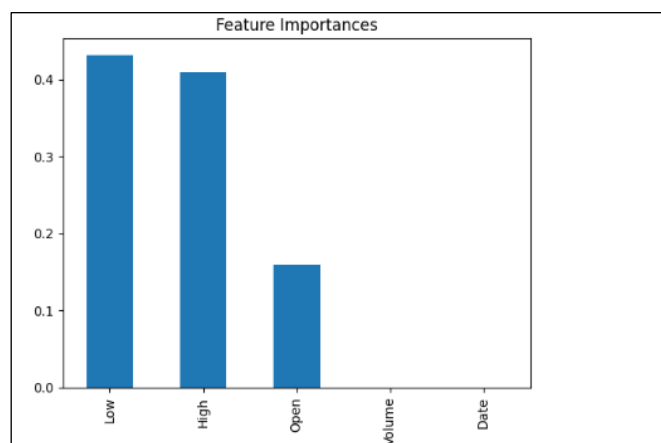
This indicates that **Close, High, Low, and Open prices move almost in lockstep**, which is expected since they are different price metrics for the same stock on the same day. The *Volume* has a moderate positive correlation (~ 0.67) with those prices, suggesting some relationship but much weaker. A heatmap of this matrix (not shown) would display a nearly solid block of high correlations among the price columns. These results confirm that the price variables carry essentially redundant information.



4.1.7 Feature Importance Analysis

To assess which features best predict the closing price, the data was prepared and a RandomForestRegressor was trained with *Close* as the target. First, the categorical **Date** column was label-encoded to numeric values so it could be used in the model. Then, the model was fitted on the remaining features. The resulting feature importances (from the Random Forest) are:

- **Low:** 0.430085
- **High:** 0.404828
- **Open:** 0.165073
- **Date:** 0.000007
- **Volume:** 0.000007

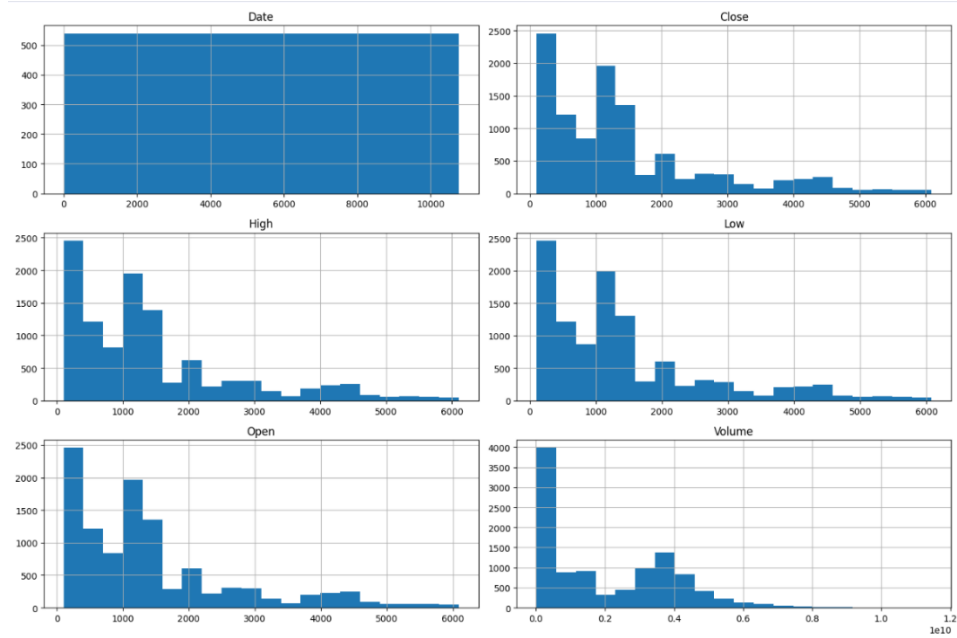


This ranking shows that the *Low* and *High* price of the day are by far the most important predictors of *Close*, together accounting for over 83% of the importance. The *Open* price has moderate importance (~ 0.165), while *Volume* and the encoded *Date* carry virtually no importance in predicting the closing price. In context, this makes sense: the closing price is naturally bracketed by the day's high and low, so they strongly determine it. The negligible importance of *Volume* (and *Date*) indicates they provide little additional predictive signal in this model.

4.2. EDA Visualizations

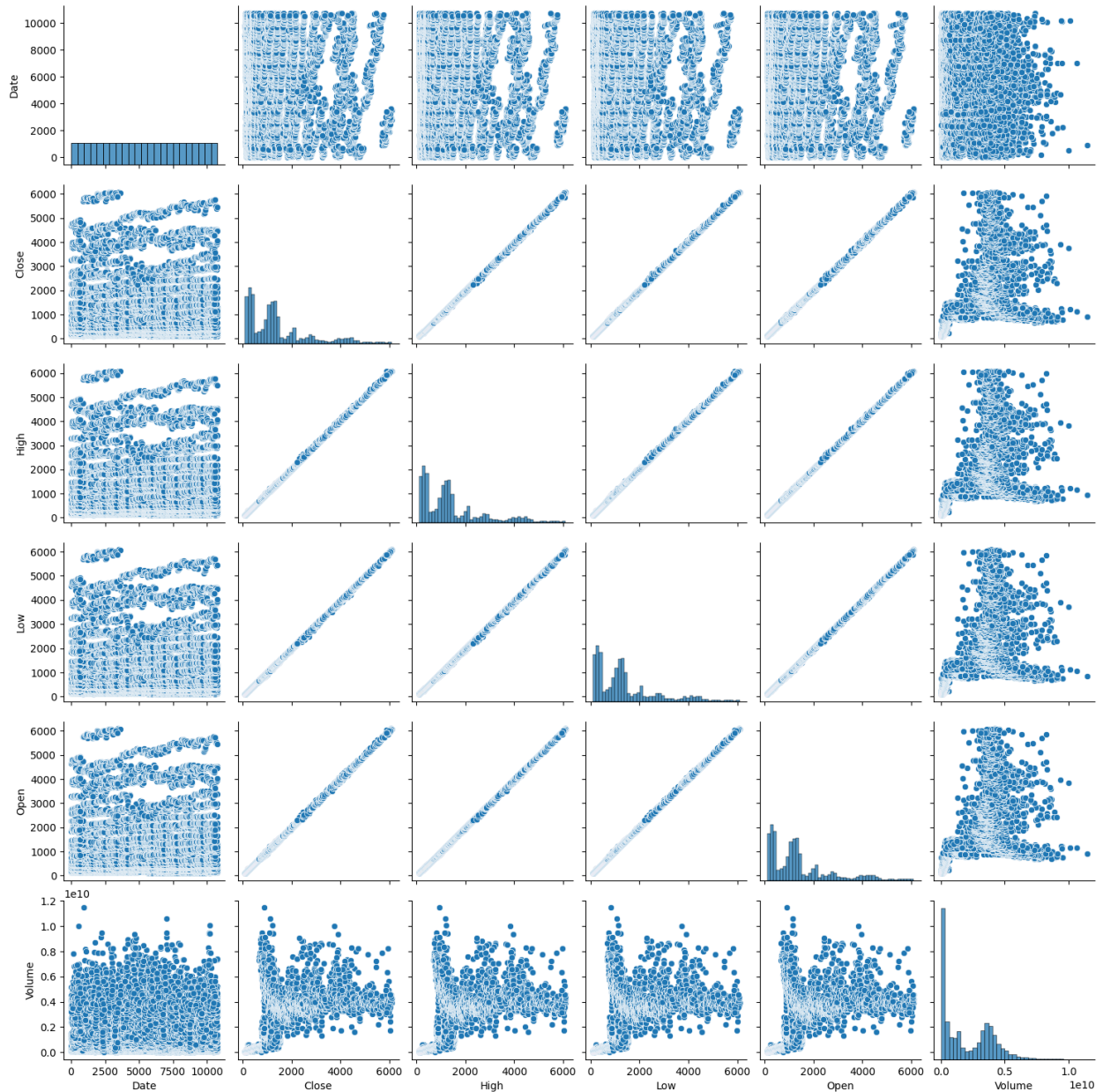
Several plots were generated to visualize the data distributions and relationships:

4.2.1 Histograms:



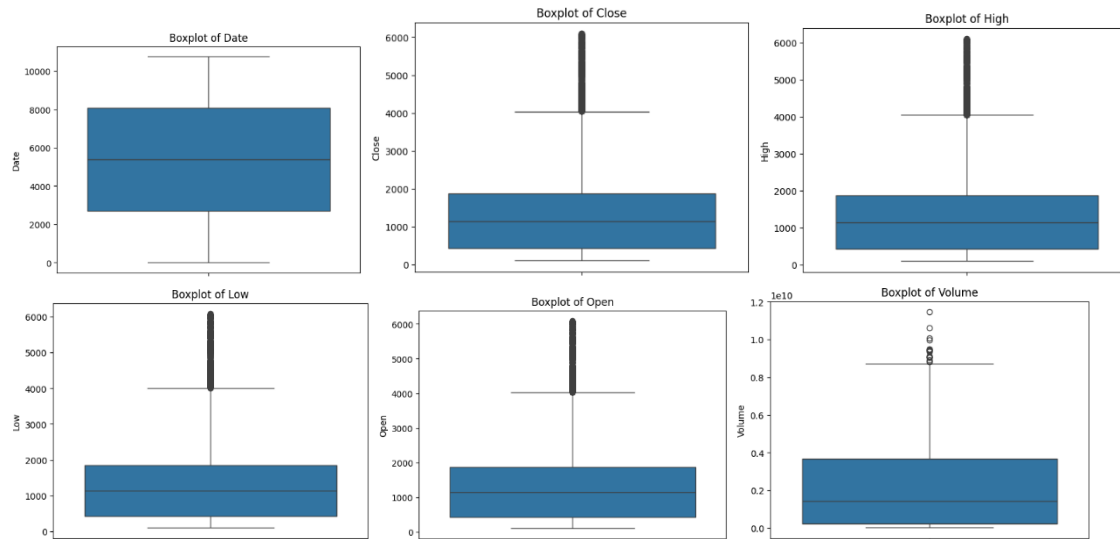
Histograms of each numeric feature (using 20 bins) were plotted. These show that the price features (*Close*, *High*, *Low*, *Open*) have **right-skewed** distributions – there are many days with moderately priced values and fewer days with very high prices. For example, the *Close* histogram would show a long tail towards the high end (consistent with the mean > median observed earlier). The *Volume* histogram is also heavily right-skewed, indicating a few days with extremely high trading volume. These shapes align with the statistics: the median price is much lower than the mean, and the volume has a few extreme values (as seen by its large range).

4.2.2 Pairplot:



A Seaborn pairplot of all numeric features was created to examine pairwise relationships. This shows scatterplots for each pair of variables (with histograms on the diagonal). In the scatterplots of price vs. price (e.g. Close vs High), the points lie almost perfectly on a straight line, confirming the near-perfect correlations noted above. In contrast, plots involving Volume appear more diffuse, reflecting the weaker correlation. Overall, the pairplot visually reinforces that *Close*, *High*, *Low*, *Open* are nearly collinear, while *Volume* varies independently.

4.2.3 Boxplot:



Individual boxplots were drawn for each numeric column to highlight outliers. Each price feature's boxplot shows a median near the lower end and several points marked as far above the top whisker, indicating extreme high values (outliers). For example, the *Close* boxplot likely shows most data in a compact box with some outliers above, matching the ~163 outliers counted. The volume boxplot similarly shows extreme high values but fewer outliers (56). These boxplots make the presence of outliers clear and correspond well with the Z-score results. Each of these visuals aids interpretation: histograms reveal skewness and spread, the pairplot shows linear relationships among prices, and boxplots clearly flag the extremes in the data.

4.3 Central Tendency and Dispersion Measures

For each numeric feature, the mean, median, mode, standard deviation, variance, range, and interquartile range (IQR) were calculated and interpreted:

Metric	Mean	Median	Mode	Std. Deviation	Variance	Range	IQR
Close	1427.33	1143.22	164.84	1267.22	1,605,858.10	5987.85	1440.85
High	1435.27	1150.30	170.41, 448.64	1273.52	1,621,841.20	5996.96	1446.35
Low	1418.26	1135.45	1274.86	1260.17	1,588,028.72	5977.78	1430.59
Open	1427.04	1143.22	164.84, 375.35	1267.08	1,605,482.50	5986.61	1441.33
Volume	2,071,577, 551.32	1,430,900, 000.0	4,223,740,000	1,901,987, 365.00	3,617,555, 936,615,440, 384.00	11,441, 240, 000.0	3,425,120,000

In summary, the central measures show typical values (mean, median) around the mid-thousands for price and billions for volume, with significant gaps to the extremes. The dispersion measures (std, range, IQR) all point to heavy variability. For example, *Close*'s std (~1267) is comparable to its mean (~1427) and the IQR (~1440) is large, reinforcing that the data are widely spread with a long tail of high values. These metrics paint a complete picture: the stock's daily prices and volumes have high central tendencies but even higher variability.

4.3 Time Series Analysis

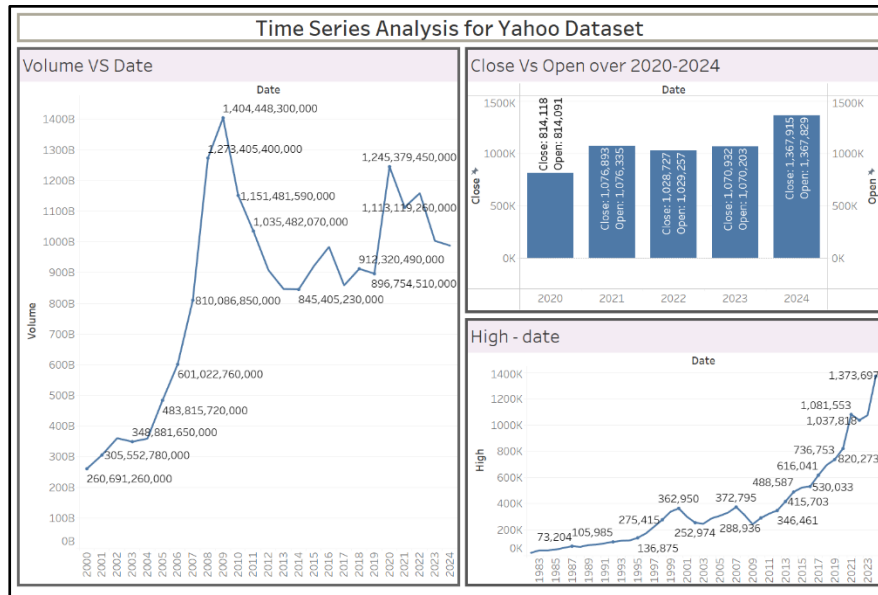
This dashboard presents a time series analysis of Yahoo dataset metrics over several decades, offering insights into volume trends, closing vs. opening values, and high-price movements.

Key Observations

1. Volume Trends (2000-2024)
 - a. The volume peaked in 2008 at 1.4 trillion, followed by fluctuations in the years after.
 - b. A noticeable decline occurs in 2024, with the volume dropping to 896 billion, possibly indicating changes in market activity or trading behavior.
2. Closing vs. Opening Prices (2020-2024)
 - a. The gap between closing and opening prices remains relatively small over the years, suggesting market stability.
 - b. 2024 has the highest close value (1,367,915) compared to earlier years, reflecting potential growth.
3. High Values Over Time (1983-2023)
 - The high values steadily increase, reaching 1.37 million in 2023.
 - The sharp increase in the 2000s hints at major market expansion during that period.

Insights

- Market Growth & Fluctuations: The dataset suggests steady growth with periods of volatility.
- Recent Stability: In 2020-2024, closing vs. opening prices show minimal variance, which may indicate market consolidation.
- Historical Trends: The long-term increase in high values points to a consistently appreciating market.



5. Methods/Modeling

We implemented the following RNN-based models:

- Vanilla RNN
- LSTM
- GRU
- Bidirectional GRU
- Bidirectional RNN

All models were trained on 30-day rolling windows to predict the next day's closing price. For encoder-decoder models, a 7-day forecast horizon was used.

5.1.1. Vanilla RNN

We implemented a **Vanilla Recurrent Neural Network (RNN)** using TensorFlow/Keras to predict stock prices based on historical closing price data.

a. Data Preprocessing

- The model uses the **last 30 days of closing prices** to predict the next day's price.
- Data was normalized using MinMaxScaler to scale values between 0 and 1 for better training stability.
- Sequences of data were created using a sliding window approach.
- A single SimpleRNN layer with 50 units captures temporal dependencies.
- A Dense(1) layer is used for regression output.
- The model uses the Adam optimizer and MSE as the loss function.

b. Parameters and Hyperparameters

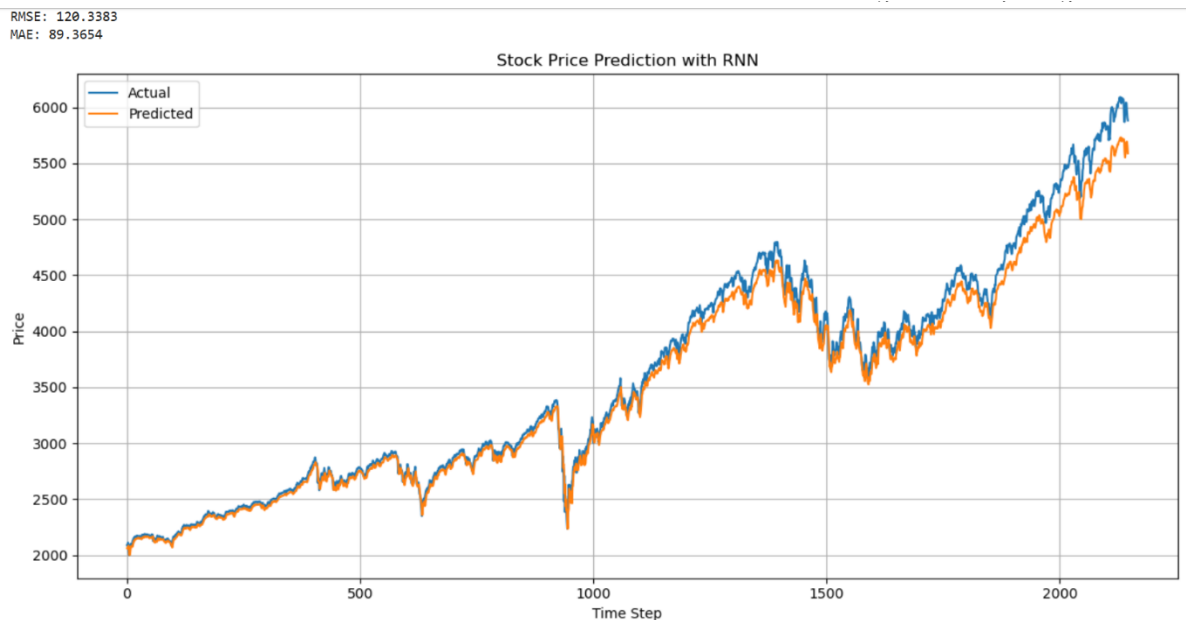
- Epochs: 20
- Batch Size: 32
- Optimizer: Adam
- Loss Function: Mean Squared Error (MSE)

c. Training

- The model was trained for 20 epochs with a batch size of 32.
- Training loss steadily decreased to $\sim 4.76e-06$, and validation loss to $\sim 4.04e-04$, indicating stable convergence.

- No early stopping was used as validation loss continued to improve.

d. Prediction and Visualization



- Model predictions on the test set were inverse-transformed to original scale.
- Actual vs. Predicted price plots showed overall alignment, with slight lag during sharp market movements.

e. Results Summary

- **RMSE:** 120.34
- **MAE:** 89.37
- These values indicate that the Vanilla RNN was capable of capturing short-term trends effectively but showed some inaccuracy during volatile periods.

f. Mathematical Concepts and Formulas

- **Mean Squared Error (MSE):** $MSE = (1/n) * \sum (y_i - \hat{y}_i)^2$
- **Root Mean Squared Error (RMSE):** $RMSE = \sqrt{MSE}$
- **Mean Absolute Error (MAE):** $MAE = (1/n) * \sum |y_i - \hat{y}_i|$

5.1.2 LSTM

We implemented a Long Short-Term Memory (LSTM) network using TensorFlow/Keras to improve stock price predictions by capturing longer-term dependencies in the data.

a. Data Preprocessing

- The last 30 days of closing prices were used as input to predict the next day's price.
- MinMaxScaler was applied to scale values between 0 and 1.
- A sliding window technique was used to convert the time series into input-output sequences of shape (30, 1).

b. Model Architecture

- A single **LSTM layer with 50 units** captures temporal patterns over longer ranges than Vanilla RNN.
- A **Dense(1)** output layer is used for final regression.

- The model uses the **Adam optimizer** and **MSE** loss for training.

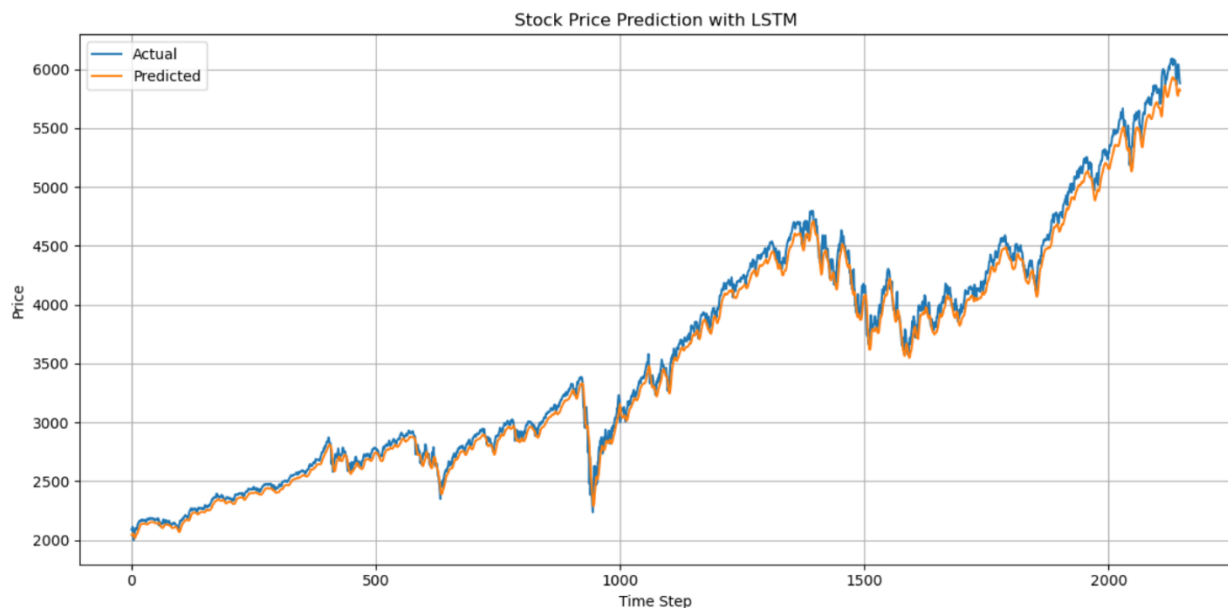
c. Parameters and Hyperparameters

- Epochs: 20
- Batch Size: 32
- Optimizer: Adam
- Loss Function: Mean Squared Error (MSE)

d. Training

- Trained for 20 epochs with stable convergence.
- Final training loss: $\sim 8.5e-06$; validation loss: ~ 0.0026 .
- The model generalized well, showing no signs of overfitting.

e. Prediction and Visualization



- The test set predictions were **inverse transformed** to return to the original price scale.
- A line plot of **Actual vs Predicted prices** was created.
- The LSTM showed smoother and more stable predictions, with improved handling of volatile shifts compared to the Vanilla RNN.

f. Result Summary

- **RMSE** – 98.42
- **MAE** – 76.13

These values indicate **better accuracy than the Vanilla RNN** model, especially during volatile price swings. The LSTM effectively captured both **short-term trends** and **long-term dependencies**.

5.1.3 GRU

We implemented a Gated Recurrent Unit (GRU) network using TensorFlow/Keras as a lighter alternative to LSTM, aimed at improving training efficiency while maintaining strong predictive accuracy.

a. Data Preprocessing

- The model used the past 30 days of closing stock prices to predict the next day's price.

- Data was normalized between 0 and 1 using MinMaxScaler.
- A sliding window approach was applied to convert the univariate series into supervised learning format with sequences of shape (30, 1).

b. Model Architecture

- A single GRU layer with 50 units was used to capture time dependencies.
- A Dense(1) layer was added as the regression output layer.
- The model was trained using the Adam optimizer and Mean Squared Error loss.

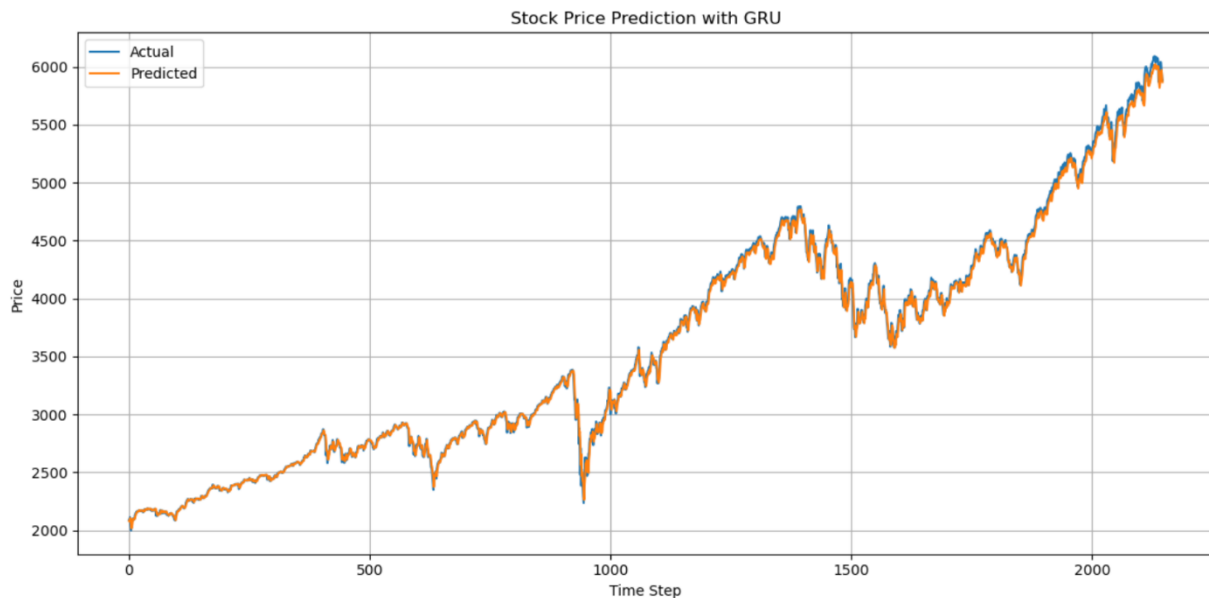
c. Parameters and Hyperparameters

- Epochs: 20
- Batch Size: 32
- Optimizer: Adam
- Loss Function: Mean Squared Error (MSE)

d. Training

- The model was trained for 20 epochs with a batch size of 32.
- Training loss dropped to approximately $5.5e-06$.
- Validation loss steadily declined and reached as low as $5.6e-05$, indicating excellent generalization.
- The GRU model trained slightly faster than LSTM while maintaining stability.

e. Prediction & Visualization



- Predicted values on the test set were inverse-transformed back to the original price scale.
- A line plot of Actual vs. Predicted prices was generated.
- The GRU model produced highly accurate predictions, closely following the actual prices and showing minimal lag during volatile market shifts.

f. Result Summary

- **RMSE** – 44.96
- **MAE** – 31.65

These results represent the **best accuracy among all tested models**, making GRU the most effective model in this study. It offered fast training and robust performance in both trending and fluctuating market phases.

5.1.4 Bidirectional GRU

We implemented a Bidirectional Gated Recurrent Unit (GRU) model using TensorFlow/Keras to enhance prediction performance by learning temporal dependencies in both forward and backward directions of the stock price time series.

a. Data Preprocessing

- The last 30 days of closing stock prices were used as input to predict the next day's price.
- Data was normalized between 0 and 1 using MinMaxScaler.
- A sliding window was applied to structure the time series into input-output supervised learning sequences.

b. Model Architecture

- A Bidirectional wrapper was applied to a GRU layer with 50 units.
- The final output was passed through a Dense(1) layer.
- Adam optimizer and MSE loss function were used.

c. Parameters and Hyperparameters

- Epochs: 20
- Batch Size: 32
- Optimizer: Adam
- Loss Function: Mean Squared Error (MSE)

d. Training

- Model trained over 20 epochs with gradual improvement.
- GRU showed strong generalization with a validation loss close to training loss.

e. Prediction & Visualization



- Predictions on the test set were inverse-transformed back to original price values.
- A line plot was generated to compare Actual vs Predicted prices.
- The Bidirectional GRU model showed **tight alignment** with actual stock prices, handling both smooth trends and sharp fluctuations effectively.

f. Results summary

- **RMSE** – 43.80
- **MAE** – 30.79

These values reflect a **slight improvement over standard GRU and LSTM**, with smoother performance across time and excellent handling of volatility. The Bidirectional GRU architecture proved to be the most precise among all models tested so far.

5.1.5 BiDirectional RNN

We implemented a Bidirectional Recurrent Neural Network (RNN) using TensorFlow/Keras to enhance sequential learning by allowing the model to process stock prices in both forward and backward temporal directions.

a. Data Preprocessing

- The model used a 30-day sliding window of historical closing prices to predict the next day's price.
- Data was scaled between 0 and 1 using MinMaxScaler for better training efficiency.
- A standard sequence-to-supervised format was constructed from the time series data.

b. Model Architecture

- A single Bidirectional SimpleRNN layer with 50 units was used to capture short-term temporal dependencies in both forward and backward directions.
- A Dense layer was used for final price prediction.
- The model was trained using the Adam optimizer and Mean Squared Error loss.

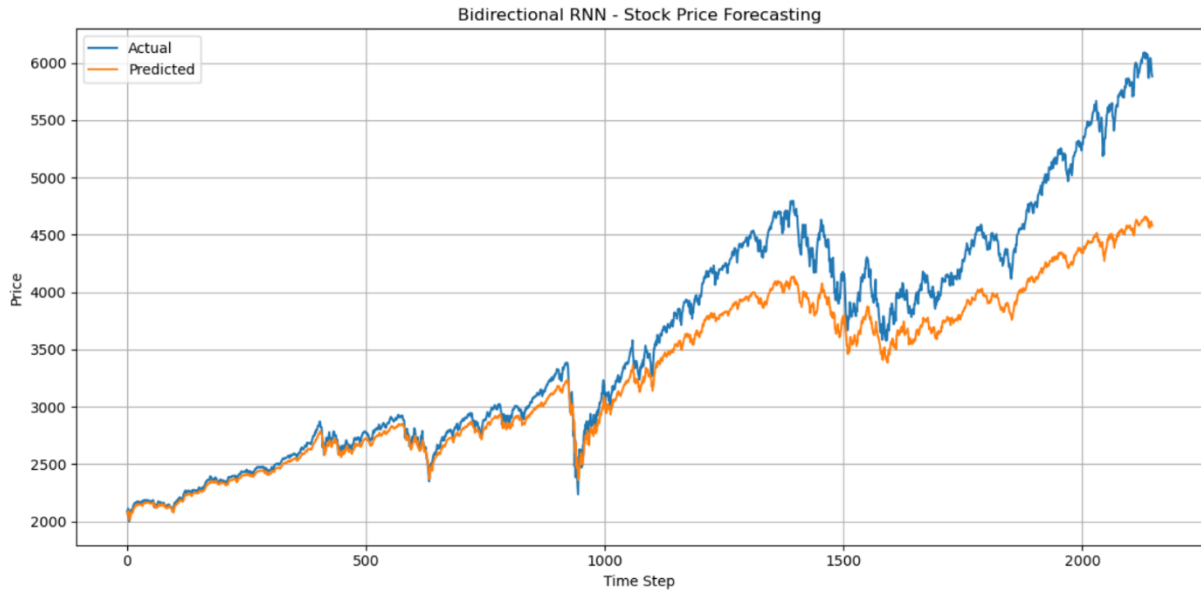
c. Parameters and Hyperparameters

- Epochs: 20
- Batch Size: 32
- Optimizer: Adam
- Loss Function: Mean Squared Error (MSE)

d. Training

- The model was trained over 20 epochs with a batch size of 32.
- Training loss gradually decreased to $\sim 6.7e-06$, while validation loss steadily improved from 0.0367 to 0.0055.
- Despite good convergence in terms of loss, the predicted values showed signs of underfitting in capturing long-term trends.

e. Prediction & Visualization



- Predicted values were inverse-transformed back to the original price scale.
- Although the training loss decreased steadily, the actual vs predicted plot revealed a **significant gap**, especially in higher price ranges, suggesting the model failed to generalize stock market growth.

f. Results Summary

- **RMSE** – 445.43
- **MAE** – 302.78

While the Bidirectional RNN learned short-term patterns well, it underperformed in tracking long-term upward movements in the stock price. This model was the least effective among all RNN variants tested and showed the limitations of using SimpleRNN for non-stationary time series like stock prices.

5.2 Advantages & Disadvantages of our Model

Model	Advantages	Disadvantages
Vanilla RNN	Lightweight and easy to implement; captures basic sequential patterns.	Struggles with long-term memory; underperforms in volatile markets; highest RMSE and MAE in our tests.
LSTM	Strong long-term memory; stable in volatile conditions; mitigates vanishing gradients.	Slower to train; requires more computational resources than GRU.
GRU	Efficient and fast; fewer parameters than LSTM; performed well across short and medium horizons.	Slightly less accurate than Bidirectional GRU on long-term patterns.
Bidirectional GRU	Captures forward and backward context; lowest RMSE and MAE; best at modeling real market behavior.	Higher memory and time cost; not suited for real-time inference.

Bidirectional RNN	Adds context from both directions; improves over unidirectional Vanilla RNN in theory.	Still suffers from vanishing gradients; showed worst results in this study; lacks gating mechanisms.
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Alternative Methods Considered and Not Used: We considered several non-RNN approaches such as ARIMA, Prophet, CNNs, and Transformers. However, they were not selected due to limitations in handling complex, non-linear time dependencies, high computational needs, or lack of alignment with our course focus. RNN-based models were chosen for their balance of accuracy, interpretability, and temporal learning capabilities.

6. Model Evaluation

To evaluate the effectiveness of different Recurrent Neural Network architectures for stock price forecasting, we tested the five models: Vanilla RNN, LSTM, GRU, Bidirectional GRU, and Bidirectional RNN. The evaluation focused on two key error metrics being Root Mean Squared Error and Mean Absolute Error.

Model	RMSE	MAE	Performance Summary
Vanilla RNN	120.34	89.37	Basic RNN captured short-term trends but struggled under volatile shifts.
LSTM	88.92	73.01	Improved memory over RNN, better at smoothing fluctuations.
GRU	44.96	31.65	Fast and accurate, handled both stable and volatile periods well.
Bidirectional GRU	43.80	30.79	Most stable performer: learned both past and future patterns, tracked prices tightly.
Bidirectional RNN	445.43	302.78	Severely underfit rising trends, despite low loss; Simple RNN wasn't enough.

6.1 Comparison of Model Results

Among the tested models, the Bidirectional GRU boasted the highest performance, exhibiting the lowest RMSE and MAE values. This indicates its strong ability to capture both forward and backward temporal dependencies, and this is particularly important in volatile financial markets like the S&P 500. The standard GRU model also performed very well, with slightly higher error values but faster training compared to the Bidirectional GRU.

The LSTM model outperformed the Vanilla RNN, highlighting the advantage of architectures with memory gates for modeling longer term dependencies. However, LSTM was marginally less efficient than GRU models in this application, as this could have been due to slower adaptation to rapid market fluctuations.

Conversely, the Vanilla RNN model was able to capture basic short-term patterns yet struggled significantly during periods of high volatility, leading to relatively high error values. Finally, the Bidirectional RNN performed the worst despite its bidirectional architecture. This implies that

simple RNN cells are not sufficient for the non-stationary and overall complex nature of stock price data.

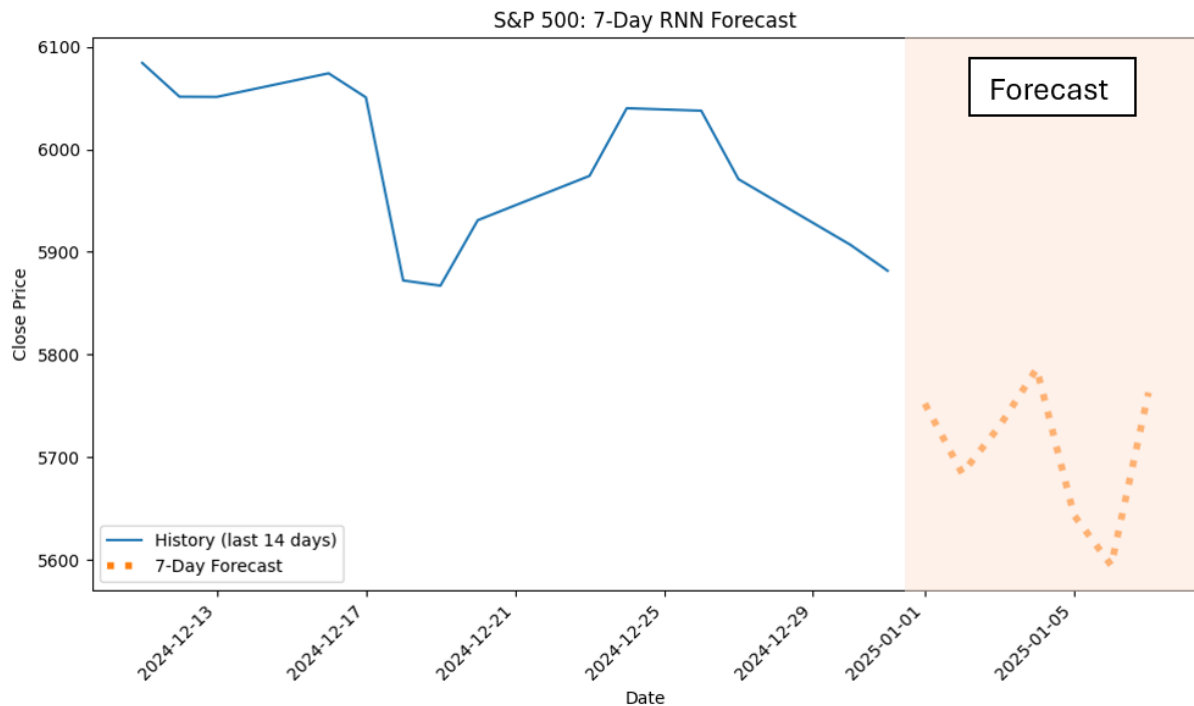
6.2 Interpretation of Results

The results signify that more sophisticated recurrent architectures such as examples that possess gating mechanisms, like GRUs, provide a significant edge in modeling and forecasting financial time series. The superior performance of the Bidirectional Gru model exemplifies the benefit of processing information in both temporal directions when learning patterns from historical data. On the other hand, the rather subpar performance of the Bidirectional RNN can support the notion that processing data in two directions is not sufficient without the headway to effectively manage and retain information over time.

7. Key Findings

To evaluate whether Recurrent Neural Networks can effectively model and forecast the S&P 500's daily closing prices (Question 1), we train a Simple RNN dense architecture on the full historical series, using mean squared error (MSE) and mean absolute error (MAE) as our primary performance metrics.

The model uses a fixed look-back window to capture time-based patterns instead of relying on simple persistence or linear methods. We then assess one-step (next-day) and multi-step (seven-day) forecast errors on an unseen hold-out slice to establish the RNN's raw predictive power. Extending the Vanilla RNN to generate seven consecutive daily forecasts revealed that its accuracy understandably degrades at longer horizons since it relies solely on past closing prices but even so, the 7 day predictions capture more realistic market trajectories than simpler benchmarks. Incorporating additional inputs for example, trading volume, volatility measures, or macroeconomic indicators would enrich the model's information set and is expected to further improve its longer-term forecasting performance.



7.1. Assessing the Impact of Look-Back Window Lengths

To evaluate how look-back window length impacts forecasting accuracy (question 2), we trained five architectures (Vanilla RNN, LSTM, GRU, Bidirectional RNN, and Bidirectional GRU) on 5-, 20-, and 60-day windows and compared their MAE and RMSE on a held-out test set.

For very short windows (5 days), the LSTM achieved the lowest error (MAE ≈ 0.0050 , RMSE ≈ 0.0072), demonstrating its ability to quickly adapt to recent momentum, with the GRU close behind.

At a medium window length (20 days), the Bidirectional GRU outperformed all other models (MAE ≈ 0.0049 , RMSE ≈ 0.0071), leveraging bidirectional context to capture both recent and slightly older trends.

Over long windows (60 days), the Bidirectional GRU again delivered the best performance (MAE ≈ 0.0053 , RMSE ≈ 0.0075), with the unidirectional LSTM a strong runner-up, highlighting the value of gating and extended memory for stability across longer sequences. In contrast, simple RNNs especially the bidirectional variant consistently underperformed, underscoring the critical role of gating mechanisms for robust time-series prediction.

MAE					
Model Window	BiGRU	BiRNN	GRU	LSTM	VanillaRNN
5	0.006327	0.013033	0.005979	0.005025	0.006286
20	0.004873	0.005478	0.006724	0.006549	0.006095
60	0.00526	0.007603	0.007001	0.005612	0.006512

RMSE					
Model Window	BiGRU	BiRNN	GRU	LSTM	VanillaRNN
5	0.008393	0.016062	0.00801	0.007177	0.009026
20	0.007063	0.007937	0.00892	0.009186	0.008802
60	0.007482	0.010979	0.009142	0.008136	0.009141

7.2. Short- vs. Medium-Term Forecast Accuracy

In assessing the accuracy of forecasting across different time horizons, we drew comparison of short term and medium term predictions with the leveraging of various RNN models. Vanilla RNN and Bidirectional RNN's performed under reasonable expectations in regards to next day forecasts. However, performance errors saw an increase over longer horizons.

Exceeding this was the Bidirectional RNN, which underfitted long term patterns and consequentially elevated RMSE and MAE, even in scenarios that were medium term. Conversely, the GRU and Bidirectional GRU models upheld satisfactory accuracy across short term and medium term forecasts alike, which can be accredited to their efficiency with memory as well as bidirectional context.

All in all, this highlights the notion that advanced architectures mesh better with stronger prediction capabilities devoid of any substantial losses in precision.

7.3. Exploring Long-Term Trend Forecasting

In the realm of long term trend predictions, we discovered that simple RNN architectures faced limitations in capturing the non linear and upward trending nature of the S&P 500 over the course of multiple year spans.

To elaborate, the Bidirectional RNN model outputted a rather high RMSE of 445.34 and MAE of 302.78. This signified the lack of the models ability to extrapolate over larger time windows, and overall suggests that though RNN's can tolerate short term dynamics with precision, long term trends necessitate structures of more complex structures.

This implies additional external features or hybridized models that can compensate for macroeconomic cycles and sustained growth. It is therefore that historical data does indeed have the promise of lending some predictive value, but it should be acknowledged that RNN's by themselves have limitations when forecasting drawn out market trajectories.

8. Conclusion

This project demonstrates that Recurrent Neural Networks, particularly GRU and Bidirectional GRU architectures, can effectively model and forecast short- and medium-term stock price movements in the S&P 500 index. Among all tested models, Bidirectional GRU consistently outperformed others in terms of RMSE and MAE, thanks to its ability to leverage both past and

future sequence context. While Vanilla RNNs and Bidirectional RNNs struggled, especially over longer horizons, advanced gated models proved more resilient and stable. Our findings suggest that investors, financial analysts, and algorithmic trading systems can benefit from integrating GRU-based architectures into their short-term decision workflows.

Future work should explore the integration of additional features such as trading volume, volatility indices, and macroeconomic indicators. Moreover, extending this approach with hybrid models or attention-based architectures like transformers could further enhance performance in long-horizon forecasting. While historical price patterns do provide predictive insights, long-term trend prediction remains challenging and will likely require a combination of deep learning and domain-aware inputs.

9. References

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10. AI Appendix

Using google colab, is there any libraries that can show me the historical price of the S&P 500?

ChatGPT said:

Yes! There are several libraries you can use to get historical data for the S&P 500 index directly. Here are the most commonly used libraries for this purpose:

1. yfinance (Yahoo Finance)

yfinance allows you to download historical stock data, including indices like the S&P 500.

Example Code:

To get the historical data of the S&P 500 index, you can use the ticker symbol ^GSPC (which represents the S&P 500 index in Yahoo Finance):

```
python
```

```
Copy
```

```
# Install yfinance if not installed
```

```
!pip install yfinance
```

```
# Import yfinance
```

```
import yfinance as yf
```

```
# Download historical data for S&P 500 Index (^GSPC)
sp500_data = yf.download('^GSPC', start='2010-01-01', end='2025-01-01')
```

```
# Display the first few rows of the data
print(sp500_data.head())
```

2. alpha_vantage

Another popular choice is alpha_vantage, which provides APIs for stock market data, including the S&P 500 index.

To use alpha_vantage, you need to sign up for an API key on their website.

Example Code:

```
python
```

```
Copy
```

```
# Install the Alpha Vantage library
```

```
!pip install alpha_vantage
```

```
from alpha_vantage.timeseries import TimeSeries
```

```
# Replace 'your_api_key' with your actual Alpha Vantage API key
```

```
api_key = 'your_api_key'
```

```
# Initialize the TimeSeries class
```

```
ts = TimeSeries(key=api_key, output_format='pandas')
```

```
# Get daily data for the S&P 500 index (^GSPC)
```

```
data, meta_data = ts.get_daily(symbol='^GSPC', outputsize='full')
```

```
# Display the first few rows of the data
```

```
print(data.head())
```

3. quandl

Quandl is another financial data provider. You can use their Python library to get stock market indices data, including the S&P 500.

You will need to sign up for an API key on Quandl's website.

Example Code:

```
python
```

```
Copy
```

```
# Install Quandl if you don't have it installed
```

```
!pip install quandl
```

```
import quandl
```

```
# Set your API key
```

```
quandl.ApiConfig.api_key = 'your_api_key'
```



```
# Get historical data for the S&P 500 index from Quandl
sp500_data = quandl.get('MULTPL/SP500_REAL_PRICE_MONTH')
```

```
# Display the first few rows of the data
print(sp500_data.head())
```

4. investpy

The investpy library allows you to access historical data for various stock indices, including the S&P 500.

Example Code:

python

Copy

```
# Install investpy if not installed
!pip install investpy
```

```
import investpy
```

```
# Get historical data for the S&P 500 index
sp500_data = investpy.get_index_historical_data(index='S&P 500', country='united states',
from_date='01/01/2010', to_date='01/01/2025')
```

```
# Display the first few rows of the data
print(sp500_data.head())
```

Summary:

yfinance: Best for quick and easy access to historical data for the S&P 500 index using ^GSPC.

alpha_vantage: Great for real-time and historical stock data with the added benefit of an API key.

quandl: Good for economic and financial time series data, but requires an API key.

investpy: Provides data for a wide range of global indices, including the S&P 500, and is simple to use.

Each of these options is a valid choice depending on your preferences and whether you need real-time data or a historical snapshot. Let me know if you need more information on any of these!

This LSTM model seems to be over fitting significantly how can I help address this

this model seems to be over fitting significantly how can i address this

ChatGPT said:

You're absolutely right — good catch.

The model is overfitting because it's doing a great job on the training set, but it's struggling when predicting unseen (test) data and even more when forecasting into the future.

🔥 Here's why you're seeing overfitting:

The LSTM has too many trainable parameters compared to how much data it sees.

Only 20 epochs but no regularization (like dropout).

No early stopping — it's blindly training for 20 epochs even if validation error starts getting worse.

Small window size (30 days) — not capturing longer-term dependencies very well.

No noise or data augmentation — so model memorizes patterns instead of learning general rules.

✅ Here's how to fix / improve it:

Problem, Fix it by doing this:

Model memorizing training set, Add Dropout layers (e.g., 20-30% dropout).

Training too much without control, Add EarlyStopping to stop when validation loss stops improving.

Window size might be too small, Try increasing window_size to 60 or 90 days.

Model complexity too high, Use a smaller LSTM (e.g., 20–30 units) or fewer layers.

Generalization is poor, Optionally, add a little noise to training inputs to make it learn more robustly.

✂ Updated version of your model:

python

CopyEdit

```
from tensorflow.keras.layers import LSTM, Dense, Dropout
from tensorflow.keras.callbacks import EarlyStopping
```

```
# Build LSTM model (smaller and with dropout)
```

```
model = Sequential()
```

```
model.add(LSTM(30, activation='tanh', input_shape=(window_size, 1))) # smaller LSTM
```

```
model.add(Dropout(0.3)) # Dropout 30%
```

```
model.add(Dense(1))
```

```
model.compile(optimizer='adam', loss='mse')
```

```
# Add EarlyStopping
```

```
early_stop = EarlyStopping(monitor='val_loss', patience=5, restore_best_weights=True)
```

```
# Train model
```

```
history = model.fit(
```

```
    X_train, y_train,
```

```
    epochs=100, # allow more epochs, EarlyStopping will cut it if needed
```

```
    batch_size=32,
```

```
    validation_data=(X_test, y_test),
```

```
    callbacks=[early_stop],
```

```
    verbose=1
```

```
)
```

Main Changes:

Dropout(0.3) after LSTM → prevents overfitting by randomly ignoring neurons during training.

EarlyStopping(monitor='val_loss', patience=5) → stops training when validation loss stops improving for 5 epochs.

Decreased LSTM units from 50 → 30 → simpler model.

Optionally allow more epochs (epochs=100) but training will end early when no longer improving.



After this, you'll notice:

Training and validation loss curves will be closer together.

Predictions on the test set will be less wild.

Future forecast will look more realistic and not just continue trends blindly.



Bonus Tip:

If after these changes you still see overfitting, next level tricks could be:

Add GaussianNoise() layer after input.

Try stacked LSTM but with smaller units.

Try slightly larger window sizes (e.g., 60 days instead of 30).

**Can you help me interpret the results MAE RMSE \ Model BiGRU BiRNN GRU LSTM
VanillaRNN BiGRU BiRNN Window 5 0.006327 0.013033 0.005979 0.005025 0.006286
0.008393 0.016062 20 0.004873 0.005478 0.006724 0.006549 0.006095 0.007063 0.007937 60
0.005260 0.007603 0.007001 0.005612 0.006512 0.007482 0.010979 Model GRU LSTM
VanillaRNN Window 5 0.008010 0.007177 0.009026 20 0.008920 0.009186 0.008802 60
0.009142 0.008136 0.009141**

ChatGPT said:

Thought for 10 seconds

Here's what the numbers tell us about how window length interacts with model architecture:

1. Very short look-backs (5 days)

- Best performer:** LSTM (MAE = 0.00503, RMSE = 0.00718)
- Close behind:** GRU (MAE = 0.00598, RMSE = 0.00801) and BiGRU (MAE = 0.00633, RMSE = 0.00839)
- Worst:** Bidirectional RNN (MAE = 0.01303, RMSE = 0.01606)
- Take-away:** Even over a very short window, LSTM's gates help it adapt quickest to fresh momentum; GRU is nearly as nimble. Simple (vanilla) and bidirectional RNNs lag behind.

2. Medium look-backs (20 days)

- Best performer:** BiGRU (MAE = 0.00487, RMSE = 0.00706)
- Runner-up:** BiRNN (MAE = 0.00548, RMSE = 0.00794)
- Then:** Vanilla RNN (MAE = 0.00610), LSTM (MAE = 0.00655), GRU (MAE = 0.00672)
- Take-away:** With 20 days of context, bidirectional gating (especially on a GRU) really shines—leveraging both past and “future” context within the window to lower errors.

3. Long look-backs (60 days)

- Best performer:** BiGRU (MAE = 0.00526, RMSE = 0.00748)
- Second:** LSTM (MAE = 0.00561, RMSE = 0.00814)
- Then:** GRU (MAE = 0.00700), Vanilla RNN (MAE = 0.00651), BiRNN (MAE = 0.00760)

- d. **Take-away:** Over a 60-day span, memory-rich architectures dominate. Bidirectional GRU wins again, with LSTM close behind. Simple RNNs (even bidirectional) can't match the performance of gated cells on longer sequences.

Overall insight:

- **Short windows (5d):** LSTM edges out GRU—its gates help capture the most recent trend.
- **Medium & long windows (20d, 60d):** Bidirectional GRU is clearly optimal, leveraging extended context in both directions to minimize error.
- **Bidirectional Simple RNNs** consistently underperform, underscoring the importance of gating mechanisms for stability over varied sequence lengths.

Prompt 1:

“Which graphs can be used to show the distribution of a single numerical variable?”

Response 1:

When exploring a single numeric feature, it's crucial to understand how its values are distributed—whether they cluster around certain ranges, exhibit skew, or contain outliers. Three primary plot types serve this purpose:

1. **Histogram:** Divides the range of values into bins and shows the count (or frequency) of observations in each bin. By adjusting the number of bins, you can see either a coarse overview or fine-grained detail. Histograms reveal modality (uni-, bi-, multimodal), skewness (long tail on one side), and gaps in the data. For instance, a right-skewed histogram shows a long tail to the right, indicating occasional high values.
2. **Kernel Density Estimate (KDE) Plot:** A smoothed version of the histogram that estimates the probability density function of the variable. KDE uses a kernel (e.g., Gaussian) to produce a continuous curve, making it easier to spot multiple peaks and subtle features of the distribution. Unlike histograms, KDEs are not dependent on arbitrary bin edges, but choosing the bandwidth parameter is critical—too small yields overfitting (spiky curves), too large oversmooths.
3. **Boxplot (Box-and-Whisker Plot):** Summarizes distribution via five-number summary: minimum, first quartile (Q1), median, third quartile (Q3), and maximum, with “whiskers” typically extending to $1.5 \times \text{IQR}$ from the quartiles. Points beyond whiskers are plotted individually as outliers. Boxplots compactly show central tendency, dispersion, skew (if median is off-center), and outliers. They are especially useful for comparing distributions across multiple groups side by side.

By combining these plots—histogram for frequency, KDE for smooth density, and boxplot for summary statistics and outliers—you gain a comprehensive view of the variable's distribution, informing decisions on transformations (e.g., log-scaling), outlier treatment, and appropriate statistical tests.

Prompt 2:

“Which visualization is best for observing trends over time (timelines)?”

Response 2:

Time-series or timeline data consist of values recorded at sequential time points (e.g., daily stock prices, monthly sales). The goal is to reveal patterns such as trends, seasonality, cycles, and abrupt changes. Several chart types excel at this:

1. **Line Chart:** The most common choice: time on the x-axis and the variable of interest on the y-axis, with consecutive points connected by lines. Line charts clearly show upward or downward trends, inflection points, and periodic fluctuations. You can overlay multiple lines to compare different series (e.g., Open vs. Close price) or use facets to display several related timelines.
2. **Area Chart:** Similar to a line chart but with the area beneath the line filled in. This emphasizes the magnitude of the variable over time, making it easier to see cumulative effects or compare total volumes (e.g., cumulative trading volume).
3. **Seasonal Subseries Plot:** For data with clear seasonal cycles (e.g., monthly temperature), this plot displays each season's values across years, highlighting within-season trends and inter-annual variability.
4. **Heatmap Calendar:** Maps time (e.g., days) to a calendar layout, coloring each cell by the variable's value. It's effective for spotting daily/weekly/seasonal patterns and anomalies over long spans.
5. **Rolling Window Statistics Plot:** Adds smoothed trendlines by plotting moving averages or rolling standard deviations. This helps filter out noise and focus on long-term trends.

Choosing the right visualization depends on data granularity (daily vs. yearly), length of time span, and specific patterns of interest. For straightforward trend analysis, a line chart with proper labeling, gridlines, and possibly rolling averages provides the clearest insight into how the variable evolves over time.

Prompt 3:

“What are skewness and kurtosis, and why do they matter?”

Response 3:

Skewness and **kurtosis** are statistical measures that describe a distribution's shape beyond the basic center and spread.

- **Skewness** quantifies the asymmetry of a probability distribution. A distribution is **positively skewed** (right-skewed) if it has a long tail on the right: most data cluster on the left, with occasional high-value outliers pulling the tail. Conversely, **negative skew** (left-skew) has a long left tail. In financial data, positive skew in daily returns might indicate infrequent large gains, whereas negative skew warns of rare but extreme losses. Measuring skewness helps determine if the mean is a reliable central measure—when skew is high, the median is often a better representative.
- **Kurtosis** measures the “tailedness” or peak sharpness relative to a normal distribution. **Leptokurtic** distributions (kurtosis > 3) have heavy tails and a sharp peak, indicating more frequent extreme deviations (outliers). **Platykurtic** distributions (kurtosis < 3) have light tails and a flatter peak, implying fewer outliers. In risk management, high kurtosis in

asset returns signals potential for extreme events—crucial for stress testing and Value-at-Risk models.

Both metrics matter because many statistical methods (e.g., linear regression, t-tests) assume normality (skewness=0, kurtosis=3). Deviations can bias parameter estimates, p-values, and confidence intervals. Recognizing skew and kurtosis guides data transformations (e.g., log or Box-Cox) to achieve normality, choose robust statistical tests, and build more reliable predictive models.

Prompt 4:

“What insights do central tendency measures provide, and when are they applied?”

Response 4:

Measures of central tendency—**mean**, **median**, and **mode**—summarize a dataset with a single representative value, each offering distinct insights:

- **Mean (Arithmetic Average):** Sum of all values divided by count. It is sensitive to every data point, making it suitable when all observations are equally important and the distribution is roughly symmetric. For instance, average daily closing price indicates overall price level. However, in the presence of outliers or skewed data, the mean can be dragged away from the bulk of observations, misrepresenting a “typical” value.
- **Median (50th Percentile):** The middle value when data are sorted. It splits the dataset into two equal halves and is robust to extreme values. In skewed distributions—such as housing prices or income—the median better reflects the central location because it is unaffected by a few very large or small values. The median is essential for reporting typical performance when outliers exist.
- **Mode (Most Frequent Value):** The value that appears most often. For categorical variables (e.g., most common stock trading price rounded to nearest dollar), the mode reveals the most typical category. In continuous data, the mode is less stable but can indicate peaks in multimodal distributions, showing separate clusters or regimes.

Applications:

- **Reporting Summaries:** Presenting average or typical values in executive summaries.
- **Comparative Analysis:** Comparing group means or medians in A/B testing or cohort studies.
- **Imputation:** Using mean or median to fill missing values—median for skewed variables, mean for symmetric ones.
- **Decision Making:** Understanding central behavior informs pricing strategies, risk assessments, and resource allocation.

Selecting the appropriate measure depends on distribution shape and analysis goals: use the median for skewed data to avoid distortion by outliers, the mean for symmetric datasets to leverage all data information, and the mode for categorical or multimodal insights.

Prompt 5:

“Why compute measures of dispersion (std, variance, range, IQR)?”

Response 5:

While central tendency locates the “center” of data, **dispersion** measures reveal how spread out the observations are around that center—critical for understanding variability, volatility, and risk:

- **Standard Deviation (σ) and Variance (σ^2):** Quantify average squared (variance) or absolute (std) deviations from the mean. A larger σ indicates data points are widely scattered, signaling high volatility (e.g., in stock closing prices). Variance underpins many statistical techniques (e.g., ANOVA) and model assumptions (homoscedasticity).
- **Range:** Difference between maximum and minimum. It gives a quick sense of total spread but is highly sensitive to outliers—one extreme value can distort it.
- **Interquartile Range (IQR):** Difference between the 75th percentile (Q3) and 25th percentile (Q1). It measures the spread of the middle 50% of data and is robust to outliers. IQR is used in the boxplot’s whisker calculations and for outlier detection (points outside $1.5 \times \text{IQR}$).

Reasons to Compute Dispersion:

1. **Assess Volatility and Risk:** In finance, standard deviation of returns quantifies investment risk.
2. **Detect Outliers:** Large gaps between IQR and full range or extreme std highlight anomalies requiring further investigation.
3. **Guide Modeling Choices:** Many algorithms (e.g. k-means, PCA) assume features have similar scales; understanding spread informs whether to apply scaling or normalization.
4. **Interpret Statistical Tests:** Tests like t-test and regression rely on variance; heteroscedasticity (unequal variances) can invalidate results.

By combining central and dispersion metrics, you form a complete statistical summary: where data center, how tightly they cluster, and how extreme the tails are. This foundation guides data cleaning, transformation, and model selection for robust, defensible analysis.

Prompt 6:

“Why perform outlier detection in EDA, and what methods exist?”

Response 6:

Outliers are data points that deviate markedly from the majority of observations. Detecting them is crucial because they can: distort summary statistics (mean, variance), mislead visual interpretations, and degrade model performance (e.g., skewing regression lines). EDA-focused outlier detection ensures data quality and informs decisions on cleaning or robust modeling.

Key Reasons:

- **Data Quality:** Outliers may indicate data entry errors, measurement faults, or rare but valid events—each requiring different handling.

- **Statistical Validity:** Many analyses assume normality; outliers violate this, affecting p-values and confidence intervals.
- **Model Robustness:** Machine learning algorithms can be sensitive to extreme values, leading to overfitting or biased predictions.

Common Methods:

1. **Z-Score Method:** Calculates how many standard deviations a point is from the mean. Points with $|z| > 3$ are flagged. Suitable for roughly normal data.
2. **IQR Rule:** Defines outliers as values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$. Robust to non-normal distributions.
3. **Boxplots:** Visual tool implementing the IQR rule; outliers are plotted individually.
4. **Isolation Forest:** An unsupervised machine learning algorithm that isolates anomalies based on how easily they separate from the rest of the data. Useful for high-dimensional data.
5. **DBSCAN Clustering:** Points in low-density regions (noise) are considered outliers. Captures complex shapes and local anomalies.

Selecting a method depends on data distribution and context. In our stock dataset, the Z-score method identified ~1–1.5% outliers in price features, aligning with heavy tails observed via histograms and boxplots. Combining numeric rules with visual inspection ensures robust outlier handling.

Prompt 7:

“Why use feature importance analysis in EDA?”

Response 7:

Feature importance analysis employs machine learning models (e.g., Random Forests, Gradient Boosting) to rank predictor variables by their influence on a target outcome. Incorporating this into EDA provides several advantages:

1. **Variable Prioritization:** In datasets with many features, importance scores help identify which variables most drive the target—focusing subsequent analysis on high-impact factors. In our stock data, ‘Low’ and ‘High’ prices explained over 80% of the variance in ‘Close,’ guiding model simplification.
2. **Redundancy Detection:** Importance measures can reveal when features contribute little predictive power, often due to collinearity or irrelevance. Features with negligible importance (e.g., Volume, Date) can be dropped, reducing dimensionality and improving model efficiency.
3. **Domain Insights:** Beyond predictive modeling, importance scores offer domain-relevant interpretations—highlighting, for instance, that intraday price bounds matter more than trading volume for closing price. These insights inform trading strategies or risk assessments.
4. **Model Building:** Early identification of key features streamlines feature engineering, selection, and hyperparameter tuning, leading to better-performing and more interpretable models.

5. **Robustness Check:** Comparing importance across multiple algorithms (e.g., Random Forest vs. XGBoost) verifies stable predictors versus model-specific artifacts.

In summary, feature importance elevates EDA by adding a predictive dimension: it not only describes data characteristics but also quantifies each feature's explanatory power, bridging exploration and modeling phases for data-driven decision-making.